ForesightNav: Anticipatory Reinforcement Learning for UAV Navigation amid Dynamic Obstacles and Large Non-Convex Structures

***Abstract*—Safe UAV navigation amid dynamic obstacles and large non-convex structures requires anticipating motion conflicts while avoiding local deadlocks induced by challenging geometry. However, existing reinforcement-learning-based policies are commonly trained using instantaneous distance-based safety signals, which can lead to late avoidance maneuvers and myopic decisions. This paper presents ForesightNav, an anticipatory reinforcement-learning framework for UAV navigation built on Proximal Policy Optimization. We introduce a finite-horizon, time-to-collision-aware three-dimensional velocity-obstacle reward that incorporates relative motion and obstacle geometry to encourage proactive avoidance of dynamic obstacles. Moreover, an action-conditioned multi-horizon auxiliary objective provides multi-step near-term outcome supervision by predicting task-relevant rollout outcomes following each observation-action pair. Its losses regularize the observation encoder shared by the actor and critic with semantically separated collision, deadlock, clearance, and progress signals. Extensive simulation experiments demonstrate that ForesightNav outperforms the compared baseline methods in dynamic-obstacle avoidance and navigation through environments containing large non-convex structures. Zero-shot real-world flights further validate its deployment feasibility without fine-tuning. These improvements require no additional online prediction or planning.**

I. Introduction

Autonomous navigation is essential for unmanned aerial vehicles (UAVs) operating amid dynamic obstacles and large non-convex structures. Beyond reacting to local geometry, a UAV must account for the likely future consequences of its decisions: an action that is collision-free at the current instant may still create a future motion conflict or steer the UAV toward a local deadlock inside geometrically confining structures. Enabling such foresight under limited onboard sensing and computation remains a central challenge in UAV navigation and control.

Classical navigation systems decompose the problem into perception, prediction, planning, and control [1]–[3]. Although interpretable and constraint-aware, these systems rely on accurate models, hand-crafted objectives, and repeated online optimization, which can increase tuning effort and reaction latency. Deep reinforcement learning offers an alternative by learning a direct observation-to-action policy from large-scale simulated experience [4]–[6]. NavRL [6] exemplifies this paradigm: it employs Proximal Policy Optimization (PPO) [7] with a reward that promotes goal-directed motion, obstacle avoidance, and regularized flight behavior. More broadly, large-structure circumnavigation is not explicitly addressed by many local reactive objectives, which emphasize immediate clearance and goal progress but do not supervise whether a sequence of locally safe actions ultimately clears an extended boundary or exits a concavity [5, 6, 8, 9]. In addition, the NavRL training objective does not explicitly model finite-horizon motion conflicts or provide horizon-specific supervision of future navigation outcomes.

These limitations manifest in two recurring failure modes. First, instantaneous distance-based safety rewards can assign similar penalties to obstacle configurations with comparable scalar separation but substantially different closing velocities, times to collision, or three-dimensional overlap geometry. Second, locally collision-free actions around large non-convex structures may still be globally unproductive. A broad wall can repeatedly block the goal direction, while L- and U-shaped structures require the policy to maintain a consistent bypass maneuver whose short-term progress may be small or temporarily negative. Without explicit guidance about these delayed outcomes, a reactive policy can oscillate, stagnate, or settle into a local optimum that appears safe at each individual step. Predictive and model-based architectures can provide temporal context or multi-step lookahead, but online rollout evaluation may increase deployment computation [10]–[12]. These limitations motivate training-time guidance that combines finite-horizon collision-risk shaping with auxiliary supervision of policy-conditioned multi-step near-term navigation outcomes.

To bridge these gaps, we extend NavRL with anticipatory training signals that address limitations of its local reactive formulation. The resulting method, ForesightNav, combines physics-informed finite-horizon collision-risk modeling with action-conditioned near-term outcome supervision to improve proactive dynamic-obstacle avoidance and reduce deadlock-prone behavior around large non-convex structures, while retaining efficient direct policy inference.

To realize this design, ForesightNav introduces two complementary training mechanisms. First, ForesightNav incorporates a finite-horizon, time-to-collision-aware reward term derived from a three-dimensional VO formulation into PPO training, encouraging the nominal policy to proactively avoid imminent motion conflicts. Second, we introduce an action-conditioned multi-horizon auxiliary prediction objective that provides multi-step near-term outcome supervision. Conditioned on the current action and the observation encoding shared by the actor and critic, the predictor estimates subsequent rollout outcomes, including future collision and deadlock events, minimum forward clearance over each prediction horizon, and cumulative progress toward the goal. The auxiliary heads provide predictive supervision only during training and are not invoked at deployment. Together, the two signals provide complementary training-time foresight at different learning levels: the TTC-aware 3D-VO reward shapes return-based policy updates using analytic relative-motion risk, whereas the action-conditioned auxiliary losses encourage the shared

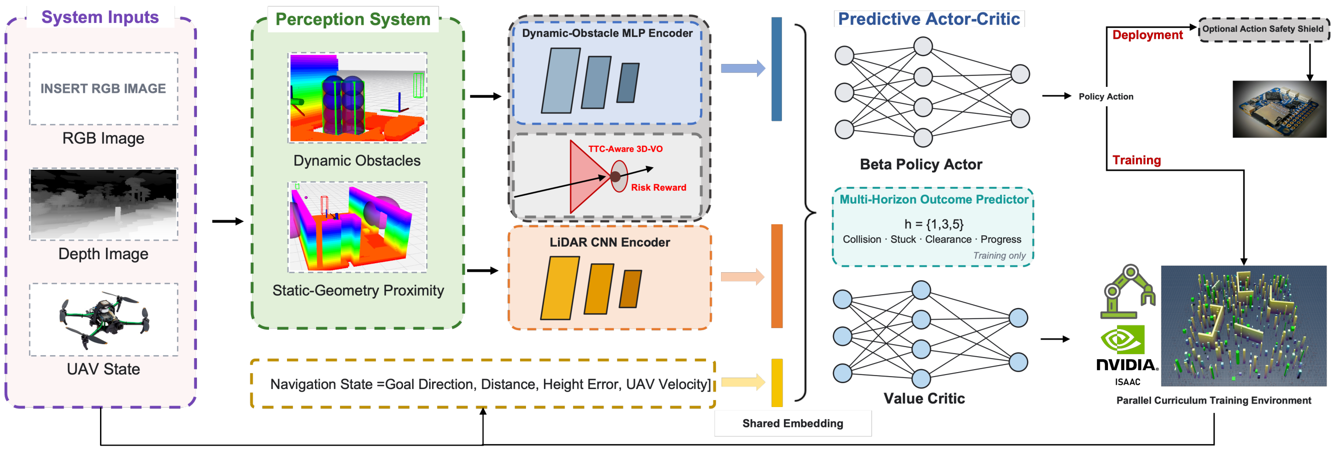


Fig. 1. The proposed ForesightNav framework. The perception system processes RGB-D observations together with the UAV state to form dynamic-obstacle, static-geometry proximity, and navigation representations. Modality-specific MLP and CNN encoders transform these inputs into a shared embedding for the predictive actor–critic. During training, the TTC-aware 3D-VO risk reward and action-conditioned multi-horizon outcome predictor jointly shape policy learning in parallel curriculum environments. During deployment, the outcome predictor is removed, and the Beta-policy actor directly generates velocity actions that can be further refined by an optional safety shield.

actor–critic encoder to retain features predictive of realized near-term outcomes following the sampled control; action selection remains governed by PPO. The overall architecture and its training and deployment pathways are illustrated in Fig. 1.

We conduct comprehensive evaluations of ForesightNav in simulation and real-world environments. In simulation, we compare it with the learning-based NavRL policy, the optimization-based PE-Planner, and an ablated variant without the 3D-VO reward [1, 6] across four settings containing static obstacles, dynamic obstacles, mixed static and dynamic obstacles, and static obstacles with large non-convex structures. ForesightNav and NavRL use identical deployment-time VO shield settings. The results demonstrate improved dynamic-obstacle avoidance and greater robustness to local trapping around large non-convex structures compared with the reference methods. Real-world UAV flights further validate the zero-shot deployment feasibility of the proposed framework.

The main contributions of this work are summarized as follows:

1) **Physics-informed 3D dynamic-risk learning:** We introduce a time-to-collision-aware three-dimensional VO reward that provides finite-horizon dynamic-risk supervision and encourages proactive avoidance of dynamic obstacles.
2) **Action-conditioned predictive representation learning:** We introduce a multi-horizon auxiliary objective that provides task-oriented supervision for the shared actor–critic encoder during training.
3) **Evaluation under dynamic and non-convex geometric challenges:** Experiments demonstrate higher navigation success around dynamic obstacles, improved deadlock and recovery metrics relative to NavRL around large non-convex structures, and zero-shot physical deployment feasibility.

II. Related Work

A. Classical, Learning-Based, and Hybrid Navigation

Classical UAV navigation uses graph search, sampling-based planning, and receding-horizon trajectory optimization. Optimization-based systems such as FASTER and ViGO generate constraint-aware trajectories through online planning and optimization [2, 3]. PE-Planner further combines kinodynamic path search and B-spline trajectory refinement with model predictive contouring control to address complex dynamic environments [1]. Learning-based methods shift computation to offline experience: CADRL learns an interaction-aware value function [4], Long *et al.* map range and goal observations to continuous controls [5], and NavRL uses PPO for direct three-dimensional UAV velocity control and applies a deployment-time shield inspired by the velocity-obstacle (VO) formulation [6]. Although discounted returns propagate future consequences, scalar reward and return signals provide only indirect and entangled supervision of distinct future collision, deadlock, clearance, and progress outcomes.

Large or non-convex structures expose a further distinction between immediate collision avoidance and successful obstacle circumnavigation. Given sufficient map coverage, a global planner can reason about connectivity and select a route around an extended boundary. In contrast, many local learning-based formulations represent only nearby geometry and optimize immediate clearance together with goal-directed motion [5, 6, 8]. They consequently provide limited explicit guidance for choosing and maintaining a bypass direction along a broad wall or for escaping an L- or U-shaped concavity, making locally safe deadlocks or local optima more likely. This challenge has motivated methods that introduce privileged route guidance; for example, Lee *et al.* use privileged time-of-arrival maps to guide quadrotors around large walls and dead ends [9].

Hybrid and safety-augmented methods rely on deployment-time planning, recovery, or action projection [13]–[17]. Other approaches use planner demonstrations or privileged expert trajectories during training [18, 19].

B. Velocity-Obstacle-Guided Learning

The VO formulation defines velocities that lead to collision under an assumed relative-motion model [20]. Classical VO methods select velocities outside these forbidden regions; ORCA distributes avoidance responsibility and solves for collision-free velocities online [21]. Used as a shield, however, VO geometry filters a nominal policy command after action generation and therefore does not train proactive avoidance.

VO structure has also shaped reinforcement-learning rewards. DRL-VO uses a two-dimensional directional reward with LiDAR histories and pedestrian kinematics [8], while Han *et al.* include RVO area and expected collision time for differential-drive robots [22]. NavRL instead applies a VO-inspired post-policy shield [6]. ForesightNav derives finite-horizon, time-to-collision-aware risk from a three-dimensional VO model, including obstacle geometry and vertical separation, to supervise short-horizon policy learning while retaining NavRL's shield as a final runtime filter.

C. Predictive Models and Auxiliary Outcome Learning

Predictive navigation methods learn representations or models of future experience. NavRep predicts latent dynamics before policy optimization [10]; RGL combines relational interaction modeling with multi-step crowd lookahead [11]; and BADGR predicts action-conditioned events for online candidate-sequence evaluation [12]. General latent models can introduce prediction error, while online rollout evaluation adds deployment computation. Auxiliary prediction can instead enrich representations while retaining direct action inference: general value functions predict temporally extended signals [23], UNREAL combines reward prediction with auxiliary control and value replay [24], and SPR learns action-conditioned multi-step latent predictions [25].

Building on NavRL, ForesightNav augments policy training with structured finite-horizon motion-risk modeling and task-oriented predictive supervision. These extensions target late responses to dynamic motion conflicts and locally safe deadlocks around large non-convex structures without adding computation to policy inference.

III. Methodology

In this section, we first introduce the finite-horizon TTC-aware 3D-VO formulation and then define the reinforcement-learning problem, including its observation space, action space, and reward function. Subsequently, we present the predictive actor–critic architecture and policy-training procedure, followed by the deployment-time safety shield.

A. Finite-Horizon TTC-Aware 3D Velocity-Obstacle Modeling

Instantaneous clearance alone cannot distinguish obstacles with similar separation but different relative motion. ForesightNav therefore extends the classical velocity-obstacle concept [20] to a finite-horizon three-dimensional formulation for quantifying dynamic motion conflicts. For a transition from $\mathbf{x}_t$ to $\mathbf{x}_{t+1}$ induced by action $\mathbf{a}_t$, the model is evaluated from the resulting state and realized UAV velocity at $t+1$.

To represent obstacle geometry without treating all interactions as planar, obstacle i, with size (w_i^x, w_i^y, h_i), is approximated by an overlapping vertical stack of spherical primitives. The sphere radius b_i, center spacing l_i, and number of spheres K_i are

$$\begin{aligned} b_i &= \frac{1}{2}\sqrt{(w_i^x)^2 + (w_i^y)^2}, \quad l_i = \max(w_i^x, w_i^y), \\ K_i &= \left\lceil \frac{h_i}{l_i} \right\rceil. \end{aligned} \tag{1}$$

For $j \in \{0, \dots, K_i - 1\}$, the center of primitive (i, j) is

$$\mathbf{p}_{i,j,t}^o = \mathbf{p}_{i,t}^o + \left[0 \quad 0 \quad \left(j - \frac{K_i - 1}{2}\right) l_i\right]^\top, \tag{2}$$

where $\mathbf{p}_{i,t}^o$ is the obstacle center. The stack is centered vertically on the original obstacle, and all primitives inherit the obstacle velocity $\mathbf{v}_{i,t}^o$.

Before collision evaluation, ForesightNav masks primitives that remain vertically irrelevant over the prediction horizon. For primitive (i, j), let $r_{i,j,t+1}^z$ and $u_{i,j,t+1}^z$ be its post-transition relative height and vertical relative velocity. The primitive is retained only if the interval bounded by $r_{i,j,t+1}^z$ and $r_{i,j,t+1}^z + H u_{i,j,t+1}^z$ intersects $[-(b_i + m_h), b_i + m_h]$, where $m_h \geq m_z$. This screening interval contains the vertical collision band and therefore does not remove a primitive solely because of a narrower prefilter.

For each retained primitive, define

$$\mathbf{r}_{i,j,t+1} = \mathbf{p}_{i,j,t+1}^o - \mathbf{p}_{t+1}, \qquad \mathbf{u}_{i,j,t+1} = \mathbf{v}_{i,t+1}^o - \mathbf{v}_{t+1}. \tag{3}$$

The inflated horizontal and vertical safety scales are $d_{i,j}^{xy} = b_i + m_{xy}$ and $d_{i,j}^z = b_i + m_z$, summarized by $\mathbf{D}_{i,j} = \mathrm{diag}(d_{i,j}^{xy}, d_{i,j}^{xy}, d_{i,j}^z)$. Under constant relative velocity, define the scaled relative trajectory

$$\mathbf{q}_{i,j,t+1}(s, \mathbf{v}) = \mathbf{D}_{i,j}^{-1}\left[\mathbf{r}_{i,j,t+1} + s(\mathbf{v}_{i,t+1}^o - \mathbf{v})\right]. \tag{4}$$

The finite-horizon velocity obstacle is then

$$\mathcal{VO}_{i,j,t+1}^H = \left\{\mathbf{v} \in \mathbb{R}^3 \;\middle|\; \exists s \in (0, H] : \|\mathbf{q}_{i,j,t+1}(s, \mathbf{v})\|_2 \leq 1\right\}. \tag{5}$$

ForesightNav evaluates whether the realized velocity $\mathbf{v}_{t+1}$ belongs to $\mathcal{VO}_{i,j,t+1}^H$, rather than explicitly constructing the full set or projecting a new velocity. Define

$$\bar{\mathbf{r}}_{i,j,t+1} = \mathbf{D}_{i,j}^{-1}\mathbf{r}_{i,j,t+1}, \qquad \bar{\mathbf{u}}_{i,j,t+1} = \mathbf{D}_{i,j}^{-1}\mathbf{u}_{i,j,t+1}. \tag{6}$$

The boundary-intersection condition is

$$A_{i,j,t+1}s^2 + B_{i,j,t+1}s + C_{i,j,t+1} = 0, \tag{7}$$

where

$$\begin{aligned} A_{i,j,t+1} &= \bar{\mathbf{u}}_{i,j,t+1}^\top \bar{\mathbf{u}}_{i,j,t+1}, \\ B_{i,j,t+1} &= 2\bar{\mathbf{r}}_{i,j,t+1}^\top \bar{\mathbf{u}}_{i,j,t+1}, \\ C_{i,j,t+1} &= \bar{\mathbf{r}}_{i,j,t+1}^\top \bar{\mathbf{r}}_{i,j,t+1} - 1. \end{aligned} \tag{8}$$

For a non-overlapping primitive, a valid time to collision exists when relative motion is nonzero and approaching, the discriminant $\Delta_{i,j,t+1} = B_{i,j,t+1}^2 - 4A_{i,j,t+1}C_{i,j,t+1}$ is nonnegative, and

$$T_{i,j,t+1} = \frac{-B_{i,j,t+1} - \sqrt{\Delta_{i,j,t+1}}}{2A_{i,j,t+1}} \in (0, H]. \tag{9}$$

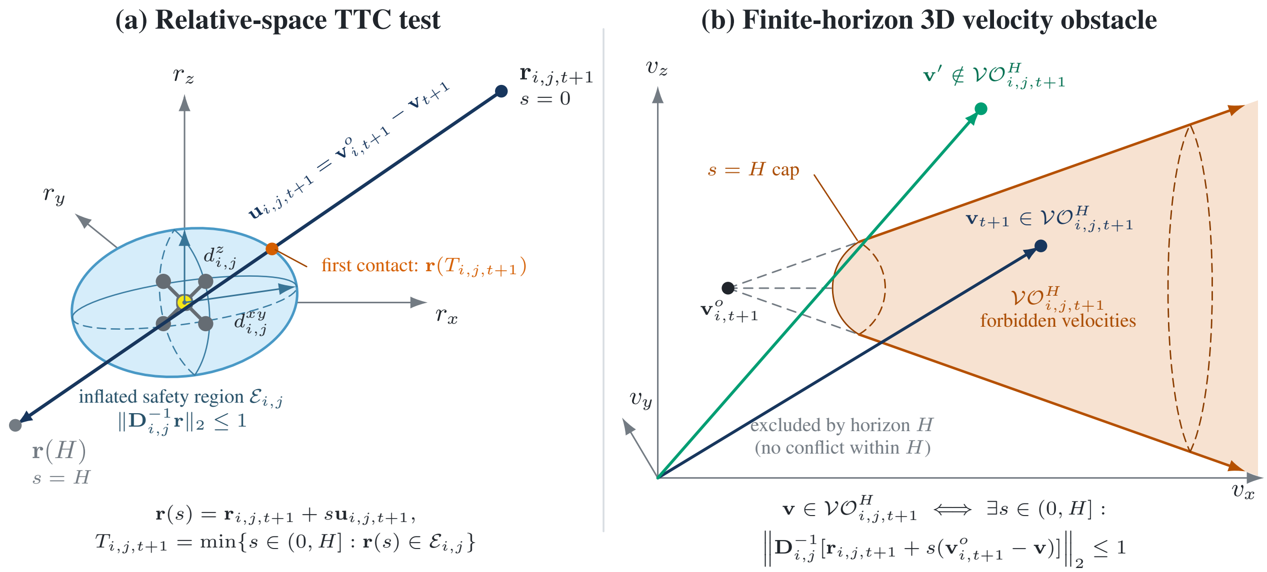


Fig. 2. Geometry of the finite-horizon TTC-aware 3D-VO model for one retained obstacle primitive. (a) The post-transition relative trajectory first intersects the anisotropically inflated safety region at $T_{i,j,t+1}$. (b) The corresponding realized UAV velocity lies inside the finite-horizon velocity obstacle; the $s = H$ cap excludes relative motions that cannot reach the safety region within the prediction horizon. The open end denotes continuation toward larger relative speeds before intersection with the bounded action space. ForesightNav evaluates this membership analytically and does not construct the set or project the policy action during training.

A primitive with $C_{i,j,t+1} \le 0$ is already inside the inflated region and is assigned unit risk. Otherwise, an in-horizon conflict receives an exponentially decaying TTC-dependent urgency score:

$$\rho_{i,j,t+1} = \begin{cases} 1, & C_{i,j,t+1} \le 0, \\ \exp(-T_{i,j,t+1}/\tau), & T_{i,j,t+1} \in (0, H], \\ 0, & \text{otherwise}. \end{cases} \quad (10)$$

The overall post-transition risk is $\rho^{\mathrm{VO}}_{t+1} = \max_{i,j} \rho_{i,j,t+1}$ over retained primitives. The maximum emphasizes the most imminent conflict, keeps the risk bounded in $[0, 1]$, and prevents its magnitude from scaling with the number of primitives. This training-time measure assumes constant relative velocity over the finite horizon and does not constitute a formal safety guarantee.

B. Reinforcement-Learning Formulation

1) Problem Formulation: ForesightNav formulates local UAV navigation as a partially observable Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \Omega, O, \gamma)$. At time t, the policy receives $\mathbf{o}_t \sim O(\cdot \mid \mathbf{x}_t)$, samples $\mathbf{a}_t \sim \pi_\theta(\cdot \mid \mathbf{o}_t)$, and induces

$$\mathbf{x}_{t+1} \sim P(\cdot \mid \mathbf{x}_t, \mathbf{a}_t), \qquad r_t = R(\mathbf{x}_t, \mathbf{a}_t, \mathbf{x}_{t+1}). \quad (11)$$

The objective is

$$\pi^\star = \arg\max_\pi \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma^t r_t \right]. \quad (12)$$

Reward and auxiliary targets may depend on latent variables excluded from the policy observation. At the beginning of each episode, a fixed navigation frame G is defined with its origin at the initial UAV position, its horizontal x-axis pointing from the initial position to the goal, and its z-axis aligned with the world vertical.

2) State and Observation Space: Following the local observation design of NavRL [6], the policy uses onboard-accessible navigation and obstacle quantities rather than the complete environment state:

$$\mathbf{o}_t = (\mathbf{L}_t, \mathbf{s}_t, \mathbf{D}_t), \quad \mathbf{L}_t \in \mathbb{R}^{36\times4}, \quad \mathbf{s}_t \in \mathbb{R}^{8}, \quad \mathbf{D}_t \in \mathbb{R}^{K_{\mathrm{dyn}}\times10}, \quad (13)$$

where $K_{\mathrm{dyn}} = 5$ is the capacity of the dynamic-obstacle observation. The LiDAR tensor contains 36 horizontal directions at 10° resolution and four elevation angles $\{-10^\circ, 0^\circ, 10^\circ, 20^\circ\}$. Entry $L_t[k, m]$ corresponds to azimuth $\psi_t + 10k^\circ$ and elevation ϕ_m, where ψ_t is the UAV yaw. The scan is indexed in this yaw-aligned sensor frame; roll and pitch are excluded from the ray orientation. A range d is encoded as the clipped proximity $R_L - d$, so larger values indicate closer geometry.

The navigation vector is

$$\mathbf{s}_t = \left[\hat{\mathbf{r}}^G_{g,t}, d^{xy}_{g,t}, \Delta z_{g,t}, \mathbf{v}^G_t\right], \quad (14)$$

where $\hat{\mathbf{r}}^G_{g,t}$ is the unit goal direction, $d^{xy}_{g,t}$ is the horizontal goal distance, $\Delta z_{g,t}$ is the vertical goal displacement, and $\mathbf{v}^G_t$ is the UAV velocity. Since G remains fixed, the lateral component of the goal direction retains signed cross-track information after the UAV departs from the initial start–goal line.

For dynamic obstacle i, let $\mathbf{r}_{i,t} = \mathbf{p}^o_{i,t} - \mathbf{p}_t$. Up to K_{dyn} in-range obstacles are ordered by increasing horizontal center distance and represented by

$$\mathbf{d}_{i,t} = \left[\hat{\mathbf{r}}^G_{i,t}, d^{xy}_{i,t}, \Delta z_{i,t}, \mathbf{v}^{o,G}_{i,t}, q_w(w_i), q_h(h_i)\right] \in \mathbb{R}^{10}. \quad (15)$$

Here, $\mathbf{v}^{o,G}_{i,t}$ is the obstacle's absolute velocity expressed in G. The width code is $q_w(w_i) = \mathrm{clip}(w_i/\delta_w - 1, 0, 3)$, with $\delta_w =$

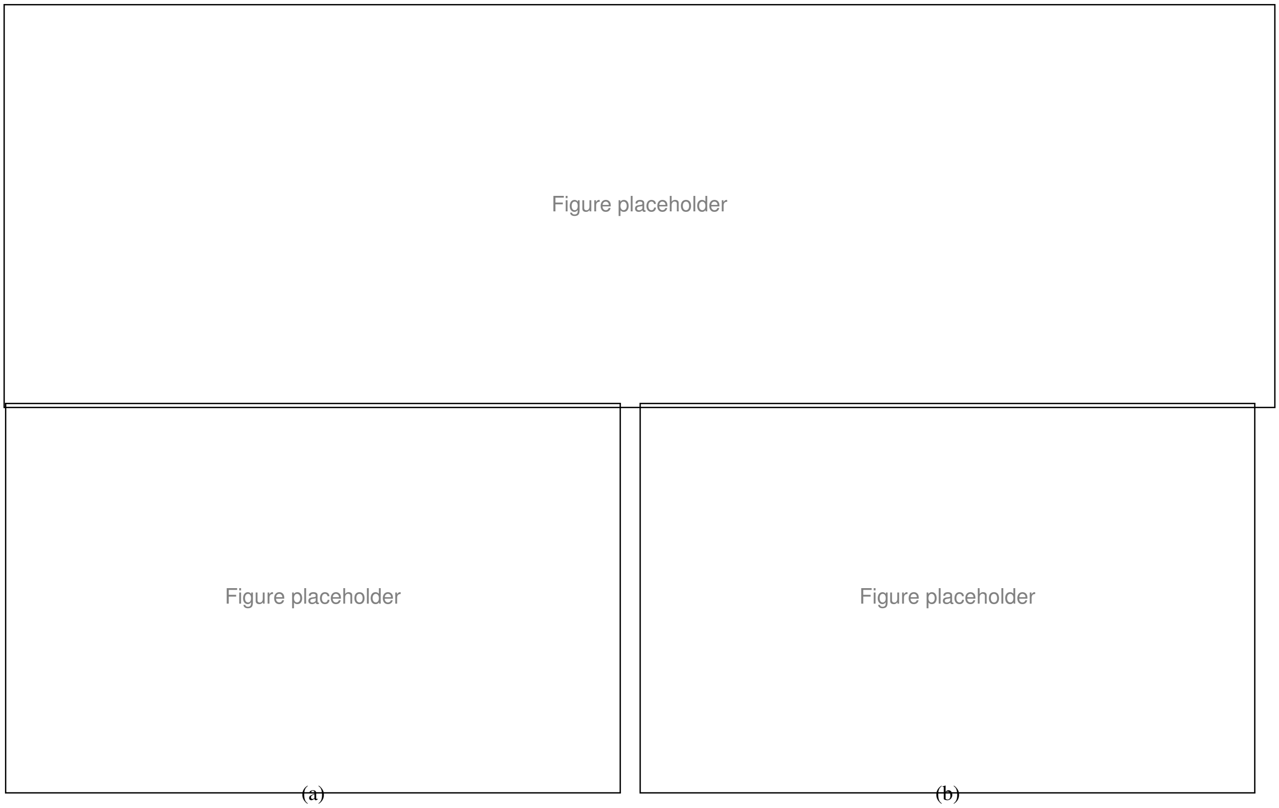


Fig. 3. Geometric illustration of the proposed goal-region amplification in the reward function. (a) Scenario without any obstacles in the reward region. (b) Scenario with obstacles entering the reward region, resulting in the generation of a penalty region.

$0.25\,\mathrm{m}$. Finite obstacles with $h_i \leq 1\,\mathrm{m}$ use $q_h(h_i) = h_i$; taller vertically spanning obstacles use $\Delta z_{i,t} = 0$ and $q_h(h_i) = 0$. Missing or out-of-range rows are zero padded. Because the encoder flattens the ordered rows, this distance ordering is part of the observation definition.

3) Action Space: Following NavRL's bounded action parameterization [6], the policy action is a three-dimensional velocity command expressed in G:

$$\mathcal{A} = [-v_{\mathrm{lim}}, v_{\mathrm{lim}}]^3. \tag{16}$$

The actor produces positive parameters $\boldsymbol{\alpha}_t, \boldsymbol{\beta}_t \in \mathbb{R}^3_{>0}$ for a factorized Beta distribution. During data collection,

$$\tilde{\mathbf{a}}_t \sim \mathrm{Beta}(\boldsymbol{\alpha}_t, \boldsymbol{\beta}_t), \qquad \mathbf{a}_t^G = v_{\mathrm{lim}}(2\tilde{\mathbf{a}}_t - \mathbf{1}), \tag{17}$$

whereas deterministic evaluation uses the distribution mean $\boldsymbol{\alpha}_t/(\boldsymbol{\alpha}_t + \boldsymbol{\beta}_t)$. The affine transformation guarantees that every sampled command satisfies the velocity limit, set to $v_{\mathrm{lim}} = 2.0\,\mathrm{m/s}$. The command is rotated into the world frame as $\mathbf{a}_t^W = \mathbf{R}_{G\to W}\mathbf{a}_t^G$ and tracked by a low-level velocity controller. Yaw is regulated separately and is not part of the learned action.

C. Reward Function

The reward combines five principal components that jointly promote anticipatory dynamic-obstacle avoidance, goal-directed motion, smooth control, static clearance, and altitude regulation:

$$r_t^{\mathrm{shape}} = \lambda_{\mathrm{VO}} r_t^{\mathrm{VO}} + \lambda_g r_t^{\mathrm{goal}} + \lambda_{\mathrm{sm}} r_t^{\mathrm{sm}} + \lambda_{\mathrm{ss}} r_t^{\mathrm{ss}} + \lambda_h r_t^h. \tag{18}$$

This combination distinguishes imminent motion conflicts from geometric proximity and balances safety with efficient, physically regular navigation.

1) TTC-Aware 3D-VO Risk Reward: The finite-horizon risk in Sec. III-A enters the reward as

$$r_t^{\mathrm{VO}} = -\eta(n)\rho_{t+1}^{\mathrm{VO}}, \qquad \eta(n) = \min\left(\frac{n}{N_{\mathrm{warm}}}, 1\right), \tag{19}$$

where n is the optimization step. The warmup factor gradually introduces anticipatory collision-risk shaping and prevents it from dominating before basic goal-directed behavior is established.

2) Goal-Progress Reward: Let $d_t = \|\mathbf{p}_g - \mathbf{p}_t\|_2$. The signed progress over the transition is

$$r_t^{\mathrm{goal}} = d_t - d_{t+1}. \tag{20}$$

Approach to the goal is rewarded, while retreat is penalized with equal magnitude. Consequently, a closed forward–backward cycle cannot obtain positive undiscounted progress reward through asymmetric shaping.

3) Smoothness Reward: Abrupt changes in realized velocity are discouraged by

$$r_t^{\mathrm{sm}} = -\|\mathbf{v}_{t+1} - \mathbf{v}_t\|_2. \tag{21}$$

This term favors temporally consistent motion and suppresses oscillatory control during local avoidance.

4) Static-Safety Reward: Let $d^L_{k,t+1}$ be the range of LiDAR ray k, N_L the number of rays, and m_s a safety margin. Static proximity is penalized by

$$r_t^{\mathrm{ss}} = -\frac{1}{N_L}\sum_{k=1}^{N_L}\frac{\max(m_s - d^L_{k,t+1}, 0)}{m_s}. \tag{22}$$

Only geometry inside the prescribed margin contributes, while averaging keeps the reward scale independent of LiDAR resolution.

5) Height Reward: To discourage unnecessary vertical detours, altitude deviation beyond a tolerance ϵ_z is penalized quadratically:

$$r_t^h = -\max(|z_{t+1} - z_g| - \epsilon_z, 0)^2. \tag{23}$$

The tolerance permits necessary altitude adjustments near obstacles, while the quadratic growth prevents avoidance through excessive ascent or descent.

D. Network Design and Policy Training

Building on NavRL's modality-specific encoder backbone [6], ForesightNav organizes policy learning around a shared predictive representation. LiDAR and dynamic-obstacle features are fused with the navigation state into $\mathbf{z}_t$. The actor generates the bounded Beta policy in Sec. III-B, while the critic estimates the observation-conditioned value $V_\psi(\mathbf{o}_t) = V_\psi(\mathbf{z}_t)$; it does not receive the latent environment state. The actor, critic, and outcome-learning pathway are optimized around this common representation.

To make $\mathbf{z}_t$ informative about the consequences of the selected action, ForesightNav introduces multi-horizon action-conditioned outcome learning. For each horizon $h \in \mathcal{H} = \{1, 3, 5\}$, a lightweight decoder predicts

$$\hat{\mathbf{y}}_t^{(h)} = g_\phi^{(h)}(\mathbf{z}_t, \tilde{\mathbf{a}}_t) = \left[\hat{c}_t^{(h)}, \hat{b}_{\mathrm{stuck},t}^{(h)}, \hat{f}_t^{(h)}, \hat{p}_t^{(h)}\right], \tag{24}$$

where $\hat{c}_t^{(h)}$ and $\hat{b}_{\mathrm{stuck},t}^{(h)}$ are collision and persistent-stuck logits, and $\hat{f}_t^{(h)}$ and $\hat{p}_t^{(h)}$ predict minimum forward clearance and cumulative goal progress. Targets are obtained from the collected transition sequence:

$$\begin{aligned} c_t^{(h)} &= \max_{1\le k\le h} c_{t+k}, & b_{\mathrm{stuck},t}^{(h)} &= \max_{1\le k\le h} b_{\mathrm{stuck},t+k}, \\ f_t^{(h)} &= \min_{1\le k\le h} f_{t+k}, & p_t^{(h)} &= \sum_{k=1}^{h}\Delta d_{t+k}, \end{aligned} \tag{25}$$

where c_t and $b_{\mathrm{stuck},t}$ are the per-step binary collision and persistent-stuck indicators, respectively, and all aggregations are truncated at episode termination. Only the current action is supplied explicitly; later actions follow the rollout policy. Hence, multi-step outputs represent outcomes of the current action under policy continuation rather than under a prescribed action sequence.

For compactness, let $\tilde{f}_t^{(h)} = \mathrm{symlog}(f_t^{(h)})$ and $\tilde{p}_t^{(h)} = \mathrm{symlog}(p_t^{(h)})$, where $\mathrm{symlog}(x) = \mathrm{sign}(x)\log(1+|x|)$. The predictive objective averages the four tasks over all horizons:

$$\begin{aligned} \mathcal{L}_{\mathrm{MH}} = \frac{1}{|\mathcal{H}|}\sum_{h\in\mathcal{H}} \big[&\ell_{\mathrm{BCE}}(\hat{c}_t^{(h)}, c_t^{(h)}) \\ &+ \ell_{\mathrm{BCE}}\Big(\hat{b}_{\mathrm{stuck},t}^{(h)}, b_{\mathrm{stuck},t}^{(h)}\Big) \\ &+ \ell_{\mathrm{SL1}}(\hat{f}_t^{(h)}, \tilde{f}_t^{(h)}) \\ &+ \ell_{\mathrm{SL1}}(\hat{p}_t^{(h)}, \tilde{p}_t^{(h)})\big]. \end{aligned} \tag{26}$$

Policy and outcome learning are integrated through

$$\mathcal{L} = \mathcal{L}_{\mathrm{PPO}} + \lambda_{\mathrm{MH}}\mathcal{L}_{\mathrm{MH}}, \qquad \lambda_{\mathrm{MH}} = 0.1. \tag{27}$$

The sampled action and rollout targets are treated as fixed inputs. Thus, $\mathcal{L}_{\mathrm{MH}}$ updates the outcome decoder and shared encoders but has no direct gradient path to the actor or critic output heads. PPO makes $\mathbf{z}_t$ useful for action selection and value estimation, while outcome supervision simultaneously makes it predictive of near-term collision, trapping, clearance, and progress.

This predictive actor–critic representation is neither recurrent nor a generative world model: it reconstructs no future observation and performs no latent rollout for action selection. The decoder is omitted during execution, while its representation-level supervision remains embedded in the shared encoder, adding no online inference cost. PPO uses clipped policy and value objectives, generalized advantage estimation, and entropy regularization [7]. A stage-wise curriculum progressively increases dynamic-interaction density and introduces more challenging large non-convex structures, strengthening both dynamic-obstacle avoidance and navigation around confining geometry while retaining previously acquired behavior.

E. Deployment-Time Policy Action Safety Shield

The learned policy provides the primary navigation command, but partial observability and unseen interactions can still yield occasional unsafe actions. ForesightNav therefore retains NavRL's optional deployment-time shield [6] as a final corrective layer. Given policy command $\mathbf{v}_{\mathrm{RL}}$, the shield passes through an admissible command and otherwise returns a nearby feasible velocity:

$$\mathbf{v}_{\mathrm{cmd}} = \begin{cases} \mathbf{v}_{\mathrm{RL}}, & \mathbf{v}_{\mathrm{RL}} \in \mathcal{V}_{\mathrm{safe}}, \\ \arg\min\limits_{\mathbf{v}\in\mathcal{V}_{\mathrm{safe}}} \|\mathbf{v} - \mathbf{v}_{\mathrm{RL}}\|_2, & \text{otherwise}, \end{cases} \tag{28}$$

where $\mathcal{V}_{\mathrm{safe}}$ is the feasible-velocity set induced by active collision-avoidance and control constraints. The shield is applied after policy inference and does not participate in policy optimization.

The training-time 3D-VO risk and deployment-time shield are complementary. TTC-aware learning encourages the nominal policy to internalize anticipatory avoidance based on obstacle geometry and relative motion. In particular, a rapidly approaching obstacle produces a shorter TTC and a stronger learning signal than a similarly positioned obstacle with weak closing motion. The policy can therefore react before a purely corrective layer has sufficient evidence or time to intervene, while the shield remains a final safeguard for residual failures rather than the sole source of dynamic-obstacle avoidance.

The evaluation consequently reports ForesightNav both without and with the shield. The former isolates the learned policy's intrinsic navigation and safety capability, whereas the latter measures the additional robustness produced by correcting residual unsafe commands. This separation prevents improvements from TTC-aware training from being conflated with deployment-time intervention.

IV. Experiment Results

In this section, we compare the proposed method with several state-of-the-art approaches in simulations across different scenarios and swarm sizes. Furthermore, we also conduct real-world experiments to validate the effectiveness and generalizability of the proposed method.

A. Training Details and Setup

The policy is trained in NVIDIA Isaac Sim using PyTorch. The training scene covers a $40\,\mathrm{m} \times 40\,\mathrm{m}$ horizontal workspace with geometry represented up to a height of $4.5\,\mathrm{m}$. The maximum UAV speed is $2.0\,\mathrm{m/s}$. To expose the policy to different obstacle scales, the radii of cylindrical dynamic obstacles are set to $\{0.125, 0.25, 0.375, 0.50\}\,\mathrm{m}$, while their speeds are sampled from 0.5 to $1.5\,\mathrm{m/s}$. Training is parallelized over $1{,}024$ UAV instances with independently randomized navigation tasks on an NVIDIA GeForce RTX 4090 GPU, and each curriculum stage requires approximately 12 hours. In addition, the hyperparameters used for training are listed in Table I; they were selected and tuned based on preliminary experiments to ensure stable and efficient training.

For quantitative deployment evaluation, we conduct 1,000 trials for each method in each scenario. A new start–goal pair is sampled for every trial, while obstacles evolve according to the motion process defined by the evaluated scenario. The environment configuration and metric thresholds are held fixed. A trial terminates upon goal arrival, collision, or the $180\,\mathrm{s}$ time limit. Let $N_{\mathrm{trial}} = 1{,}000$ be the number of trials, and let N_{succ}, N_{coll}, and N_{timeout} denote the numbers of successful, collided, and timed-out trials, respectively. We further denote by $N^{\mathrm{evt}}_{\mathrm{dead}}$ the total number of deadlock events detected across all trials and by $N^{\mathrm{evt}}_{\mathrm{rec}}$ the recovered-deadlock-event count (`recovered_deadlock_event_count`). We report the following metrics:

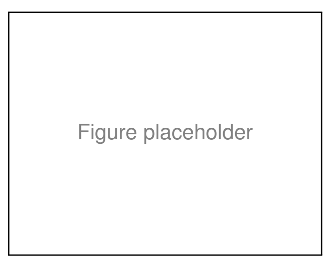
- *Success Rate* ($N_{\mathrm{succ}}/N_{\mathrm{trial}}$): The fraction of trials in which the UAV reaches the goal before collision or timeout.
- *Collision Rate* ($N_{\mathrm{coll}}/N_{\mathrm{trial}}$): The fraction of trials terminated by contact with a static or dynamic obstacle.
- *Timeout Rate* ($N_{\mathrm{timeout}}/N_{\mathrm{trial}}$): The fraction of trials that neither reach the goal nor collide within $180\,\mathrm{s}$.
- *Deadlock Count* ($N^{\mathrm{evt}}_{\mathrm{dead}}$): The total number of deadlock events detected over the 1,000 trials. A deadlock is registered when a forward obstacle co-occurs with goal progress no greater than $0.005\,\mathrm{m}$ for 40 consecutive evaluation cycles ($2\,\mathrm{s}$ at $20\,\mathrm{Hz}$). Because one trial may contain multiple events, this count is not bounded by N_{trial}.
- *Post-Deadlock Escape Rate* ($N^{\mathrm{evt}}_{\mathrm{rec}}/N^{\mathrm{evt}}_{\mathrm{dead}}$): The fraction of detected deadlock events that satisfy the recovery criterion. This event-level metric is distinct from the fraction of deadlocked trials that eventually reach the goal and is reported as undefined when no deadlock event occurs.

TABLE I
KEY HYPERPARAMETERS USED FOR POLICY TRAINING

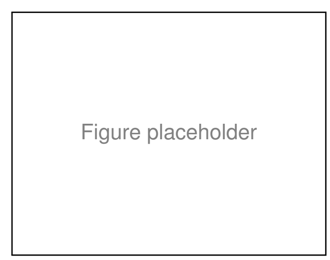
Parameter	**Value**	**Parameter**	**Value**
Parallel UAVs	1,024	Rollout length	32
PPO epochs	4	Minibatches	16
Discount factor γ	0.99	GAE parameter λ_{GAE}	0.95
Initial LRs (encoder/actor/critic)	$(5,5,5) \times 10^{-4}$	Adaptation LRs (encoder/actor/critic)	$10^{-5}, 10^{-5}, 10^{-4}$
Prediction horizons $\mathcal{H}$	$\{1,3,5\}$	Outcome-loss weight λ_{MH}	0.1
VO horizon H_{VO}	$2.1\,\mathrm{s}$	TTC decay τ	$0.75\,\mathrm{s}$
VO reward weight λ_{VO}	1.3	VO warmup	20,000 steps

Taken together, these metrics characterize the UAV's ability to avoid dynamic obstacles and navigate around large non-convex structures, including its tendency to avoid or recover from deadlocks and local optima. Comparisons with the corresponding ablations further assess whether TTC-aware and multi-horizon prediction enables earlier responses to evolving motion conflicts.

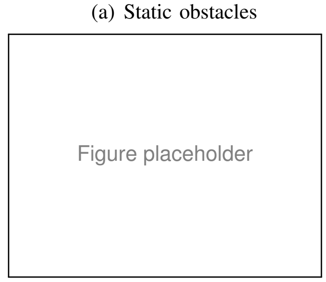
B. Simulation Results



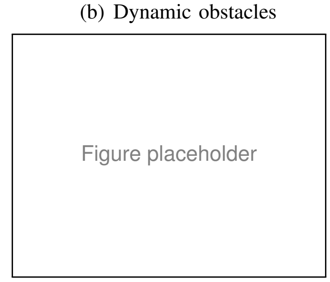
(a) Static obstacles



(b) Dynamic obstacles



(c) Mixed obstacles



(d) Large non-convex structures

Fig. 4. Representative ForesightNav trajectories in the four simulation environments.

We conducted simulation experiments in Gazebo to evaluate ForesightNav and its sim-to-sim transferability from the training environment. The quantitative evaluation covers four challenging settings containing static obstacles, dynamic obstacles, mixed obstacles, and large non-convex structures.

TABLE II
QUANTITATIVE COMPARISON OVER 1,000 TRIALS IN FOUR OBSTACLE ENVIRONMENTS WITHIN A $20\,\mathrm{m} \times 20\,\mathrm{m}$ WORKSPACE.

Environment	**Method**	**Success Rate (%) ↑**	**Collision Rate (%) ↓**	**Timeout Rate (%) ↓**	**Deadlock Count ↓**	**Post-Deadlock Escape Rate (%) ↑**
Env. 1: Static obstacles ($N_{\mathrm{stat}} = 70$)	PE-Planner	85.5	14.5	0	0	N/A
	NavRL	94.1	5.9	0	0	N/A
	ForesightNav w/o 3D-VO	92.7	7.3	0	0	N/A
	ForesightNav	94.4	5.6	0	0	N/A
Env. 2: Dynamic obstacles ($N_{\mathrm{dyn}} = 60$)	PE-Planner	65.6	34.4	0	0	N/A
	NavRL	69.9	30.1	0	0	N/A
	ForesightNav w/o 3D-VO	71.2	28.8	0	0	N/A
	ForesightNav	83.3	16.7	0	0	N/A
Env. 3: Mixed static and dynamic obstacles ($N_{\mathrm{stat}} = 20$) ($N_{\mathrm{dyn}} = 40$)	PE-Planner	63.2	36.8	0	0	N/A
	NavRL	61.8	38.2	0	0	N/A
	ForesightNav w/o 3D-VO	60.3	39.7	0	0	N/A
	ForesightNav	76.5	23.5	0	0	N/A
Env. 4: Static obstacles and large non-convex structures	PE-Planner	58.9	15.9	25.2	127	21/127 (16.5%)
	NavRL	66.7	13.6	19.7	93	23/93 (24.7%)
	ForesightNav w/o 3D-VO	77.5	14.0	8.5	42	18/42 (42.9%)
	ForesightNav	78.1	15.6	6.3	53	24/53 (45.3%)

We compare ForesightNav with the learning-based NavRL policy [6], the optimization-based PE-Planner [1], and an ablation without the TTC-aware 3D-VO reward. Each method is evaluated over 1,000 independent trials under the same workspace, start–goal sampling, obstacle configuration, and termination criteria. Representative trajectories are shown in Fig. 4, and the quantitative results are reported in Table II.

ForesightNav maintains performance comparable to NavRL in the static environment and provides a clearer advantage when obstacle motion must be anticipated. In the purely dynamic environment, its success rate reaches 83.3%, compared with 69.9% for NavRL. The same overall trend is observed in the mixed environment, where ForesightNav achieves the highest success rate and the lowest collision rate. The degradation caused by removing the 3D-VO reward further indicates that finite-horizon relative-motion supervision improves responses to evolving motion conflicts rather than merely increasing static clearance.

In the environment containing large non-convex structures, both ForesightNav variants reduce timeouts and deadlock-related failures relative to NavRL and PE-Planner, while recovering a larger proportion of detected deadlocks. The complete method and the 3D-VO ablation exhibit mixed differences on individual metrics; therefore, this environment demonstrates the benefit of the broader predictive training design under confining geometry rather than an isolated advantage of the 3D-VO component. Overall, the results show that ForesightNav preserves static-obstacle navigation while improving anticipatory dynamic avoidance and robustness to local trapping around large structures.

Table III further evaluates the deployment-time safety shield in the mixed environment. Enabling the shield consistently increases the success rate and reduces collisions for both NavRL and ForesightNav, with a larger improvement observed for ForesightNav. This result confirms that learned anticipatory avoidance and online action correction are complementary: the policy provides the nominal navigation behavior, while the shield mitigates residual unsafe commands.

TABLE III
EFFECT OF THE DEPLOYMENT-TIME SAFETY SHIELD OVER 1,000 TRIALS IN ENVIRONMENT 3 (MIXED STATIC AND DYNAMIC OBSTACLES) WITHIN A $20\,\mathrm{m} \times 20\,\mathrm{m}$ WORKSPACE.

Method	**Shield**	**Success Rate (%) ↑**	**Collision Rate (%) ↓**	**Timeout Rate (%) ↓**
NavRL	Off	57.4	42.6	N/A
	On	62.7	37.3	N/A
ForesightNav	Off	64.2	35.8	N/A
	On	75.3	24.7	N/A

C. Physical Flight Tests

To evaluate zero-shot sim-to-real transfer, we deploy the policy selected in simulation on a quadrotor without additional fine-tuning. The experiments are conducted in a $28\,\mathrm{m} \times 15\,\mathrm{m}$ flight area, and the maximum translational velocity is set to $1.2\,\mathrm{m/s}$. The onboard navigation stack supplies the policy with the same local observation schema used during training, comprising proximity measurements, the estimated UAV navigation state, and tracked dynamic-obstacle descriptors. ForesightNav produces three-dimensional velocity commands that are tracked by the low-level flight controller. The multi-horizon outcome predictor and TTC-aware 3D-VO risk computation are used only for training and are not evaluated onboard.

Figure placeholder

(a) Static

Figure placeholder

(b) Dynamic

Figure placeholder

(c) Mixed

Figure placeholder

(d) Non-convex

Figure placeholder

(e) Linear velocity

Fig. 5. Zero-shot physical flight tests of ForesightNav in a $28\,\mathrm{m} \times 15\,\mathrm{m}$ area. (a) Static obstacles. (b) Dynamic obstacles. (c) Mixed static and dynamic obstacles. (d) Large non-convex structures. (e) UAV linear-velocity profiles in the four environments.

We construct four physical environments to examine complementary navigation capabilities. As shown in Fig. 5(a)–(d), the first environment contains static obstacles, the second introduces moving obstacles, the third combines static and moving obstacles, and the fourth contains large non-convex structures. These settings respectively examine local clearance maintenance, response to relative obstacle motion, avoidance under simultaneous geometric and motion constraints, and sustained detours around structures that can induce local trapping. The four layouts therefore provide a progressively broader assessment than a single open-space avoidance test.

Figure 5(e) complements the flight scenes by reporting the evolution of UAV linear velocity in all four environments under the common $1.2\,\mathrm{m/s}$ limit. The profiles expose both the acceleration required for goal-directed flight and the speed adjustments made during obstacle interactions, rather than evaluating transferability from final paths alone. Together, the representative flights and velocity histories demonstrate closed-loop operation of the transferred policy under real sensing and control effects. These tests provide qualitative evidence of practical deployment feasibility, while the controlled statistical comparisons remain those reported in the simulation study.

V. Conclusion

This paper presented ForesightNav, an anticipatory reinforcement-learning framework that extends NavRL for UAV navigation amid dynamic obstacles and large non-convex structures. A finite-horizon TTC-aware 3D-VO reward introduces relative-motion risk into policy learning, while action-conditioned multi-horizon outcome supervision makes the shared actor–critic representation predictive of near-term collision, trapping, clearance, and progress. Both mechanisms are confined to training, preserving direct policy inference at deployment. Simulation results show improved dynamic-obstacle avoidance and greater robustness to timeouts and local trapping around large structures, and zero-shot flights demonstrate practical deployment feasibility. The formulation remains limited by local observations and a constant-relative-velocity assumption and does not provide a formal safety guarantee. Future work will investigate uncertainty-aware motion prediction and compact temporal representations for longer-horizon navigation around partially observed structures.

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